

Practical 4

About this unit

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Practice Lab Assignment

Unit • 100% completed



Lab Assignment

Unit • 100% completed



4.1.1. Pandas - series creation and manipulation

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the `groupby` and `mean()` operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

Sample Test Cases

seriesMa...

```
1 import pandas as pd
2
3 # Take inputs from the user to create a list of numbers
4 numbers = list(map(int, input().split()))
5
6 # Create a Pandas series from the list of numbers
7 series = pd.Series(numbers)
```

Average time: 0.005 s, 5.17 ms | Maximum time: 0.015 s, 15.00 ms | 3 out of 3 shown test case(s) passed | 3 out of 3 hidden test case(s) passed

Test case 1 (5 ms) | Debug | [Icons]

Expected output	Actual output
1 2 3 4 5 6 7 8 9 10	1 2 3 4 5 6 7 8 9 10
Mean of even and odd numbers:	Mean of even and odd numbers:
Odd: 5.0	Odd: 5.0
Even: 6.0	Even: 6.0
dtype: float64	dtype: float64

1.2. Dictionary to dataframe

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

datafram...

```
1 import pandas as pd
2
3 # Provided dictionary of lists
4 data = {
5     'Name': ['Alice', 'Bob', 'Charlie'],
6     'Age': [25, 30, 35],
7 }
```

Average time: 0.036 s, 36.50 ms | Maximum time: 0.043 s, 43.00 ms | 1 out of 1 shown test case(s) passed | 1 out of 1 hidden test case(s) passed

Test case 1 (43 ms) | Debug | [Icons]

Expected output	Actual output
Original DataFrame:	Original DataFrame:
.....Name.....Age.....	Name.....Age.....
0.....Alice.....25.....	0.....Alice.....25.....
1.....Bob.....30.....	1.....Bob.....30.....
2.....Charlie.....35.....	2.....Charlie.....35.....
New name: Susan	New name: Susan
New age: 25	New age: 25
After adding a row:	After adding a row:

1.3. Student Information

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:

Refer to the displayed test cases for better understanding.

Sample Test Cases

studentin...

```
1 import pandas as pd
2
3 # Read the text file into a DataFrame
4 file = input()
5 data = pd.read_csv(file, sep="\\s+", header=None, names=["Name", "Age", "Grade"])
6
7 print("First five rows:")
```

Average time: 0.023 s, 23.00 ms | Maximum time: 0.035 s, 35.00 ms | 1 out of 1 shown test case(s) passed | 1 out of 1 hidden test case(s) passed

Test case 1 (35 ms) | Debug | [Icons]

Expected output	Actual output
studentdata.txt	studentdata.txt
First five rows:	First five rows:
.....Name.....Age.....Grade.....	Name.....Age.....Grade.....
0.....John.....25.....A.....	0.....John.....25.....A.....
1.....Ali.....22.....B.....	1.....Ali.....22.....B.....
2.....Emma.....24.....A.....	2.....Emma.....24.....A.....

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4.2.1. Month with the Highest Total Sales

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8 df["Date"] = pd.to_datetime(df["Date"])
```

Average time0.014 sMaximum time14.33 ms

1 out of 1 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 1

Expected outputsales_data.csvActual outputsales_data.csv

Best month: 2025-01Best month: 2025-01

Total sales: \$12210.00Total sales: \$12210.00

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4.2.3. City that Sold the Most Products

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by city and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
```

Average time0.010 sMaximum time10.00 ms

1 out of 1 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 1

Expected outputsales_data.csvActual outputsales_data.csv

City sold the most products: Los AngelesCity sold the most products: Los Angeles

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4.2.2. Best Selling Product

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
```

Average time0.010 sMaximum time10.00 ms

1 out of 1 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 1

Expected outputsales_data.csvActual outputsales_data.csv

Best selling product: Product ABest selling product: Product A

Total quantity sold: 23Total quantity sold: 23

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4.2.3. City that Sold the Most Products

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by city and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
```

Average time0.010 sMaximum time10.00 ms

1 out of 1 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 1

Expected outputsales_data.csvActual outputsales_data.csv

City sold the most products: Los AngelesCity sold the most products: Los Angeles

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4.2.3. City that Sold the Most Products

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by city and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
```

Average time0.010 sMaximum time10.00 ms

1 out of 1 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 1

Expected outputsales_data.csvActual outputsales_data.csv

City sold the most products: Los AngelesCity sold the most products: Los Angeles

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4.2.4. Most Frequently Sold Product Pairs

100%

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pairs that was sold most frequently.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Explanation:

Transactions:

- 2025-01-01: Product A, Product B

Explorer

frequent...

Submit

1import pandas as pd

2from itertools import combinations

3from collections import Counter

4

5# Prompt user to input the file name

Average time0.010 s10.33 ms

Maximum time0.021 s21.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 141 msDebug

Expected output

Actual output

sales_data.csv

Product A and Product B: 2 times

Product A and Product B: 2 times

Product A and Product C: 2 times

Product A and Product C: 2 times

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4.2.5. Titanic Dataset Analysis and Data Cleaning

100%

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

- Display the first 5 rows of the dataset.
- Display the last 5 rows of the dataset.
- Get the shape of the dataset (number of rows and columns).
- Get a summary of the dataset (using `.info()`).
- Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
- Check for missing values and display the count of missing values for each column.
- Fill missing values in the 'Age' column with the median age.
- Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
- Drop the 'Cabin' column due to many missing values.

1. Display the first 5 rows of the dataset.

2. Display the last 5 rows of the dataset.

3. Get the shape of the dataset (number of rows and columns).

4. Get a summary of the dataset (using `.info()`).

5. Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.

6. Check for missing values and display the count of missing values for each column.

7. Fill missing values in the 'Age' column with the median age.

8. Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).

9. Drop the 'Cabin' column due to many missing values.

Explorer

titanicDat...

Submit

1import pandas as pd

2import numpy as np

3

4# Load the Titanic dataset

5data = pd.read_csv('Titanic-Dataset.csv')

6

Average time0.092 s92.00 ms

Maximum time0.092 s92.00 ms

1 out of 1 shown test case(s) passed

Test case 142 msDebug

Expected output

Actual output

PassengerId Survived Pclass Fare Cabin Embarked

PassengerId Survived Pclass Fare Cabin Embarked

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4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

100%

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
- Convert the 'Sex' column to numeric values (male: 0, female: 1).
- One-hot encode the 'Embarked' column, dropping the first category.
- Get the mean age of passengers.
- Get the median fare of passengers.
- Get the number of passengers by class.
- Get the number of passengers by gender.
- Get the number of passengers by survival status.
- Calculate the survival rate of passengers.
- Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

Explorer

titanicDat...

Submit

1import pandas as pd

2import numpy as np

3

4# Load the Titanic dataset

Average time0.040 s40.00 ms

Maximum time0.040 s40.00 ms

1 out of 1 shown test case(s) passed

Test case 140 msDebug

Expected output

Actual output

29.69911764795882

29.69911764795882

14.4542

14.4542

3 ... 491

3 ... 491